Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A battery of emf E and internal resistance r is connected across a pure resistive device (such as an electric heater) of resistance R. The power output of the device will be maximum if

(A) $R = r$

(B) $R = \sqrt{2}r$

(C) $R = 2r$

(D) $R = 4r$

Q.2

An electric kettle has two coils. When one coil is connected to the ac mains, the water in the kettle boils in 10 minutes. When the other coil is used, the same quantity of water takes 15 minutes to boil. How long will it take for the same quantity of water to boil if the two coils are connected in parallel?

(A) 6 min

(B) 12 min

(C) 18 min

(D) 24 min

Q.3

You are given a resistance coil and a battery. In which of the following cases is largest amount of heat generated?

(A) When the coil is connected to the battery directly

(B) When the coil is divided into two equal parts and both the parts are connected to the battery in parallel

(C) When the coil is divided into four equal parts and all the four parts are connected to the battery in parallel

(D) When only half the coil is connected to the battery

Q.4

If voltage across a bulb rated 220 Volt-100 Watt drops by 2.5% of its rated value, the percentage of the rated value by which the power would decrease is :

(A) 20%

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(B) 2.5%

(C) 5%

(D) 10%

Q.5

A battery is charged at a potential of $15\ \mathrm{V}$ for 8 hours when the current flowing is $10\ \mathrm{A}$. The battery on discharge supplies a current of 5 A for 15 hours. The mean terminal voltage during discharge is $14\ \mathrm{V}$. The "watt-hour" efficiency (in percent) of the battery is

(A) 87.5%

(B) 75.5%

(C) 95%

(D) 90%

Q.6

A electric tea kettle has two heating coils. When first coil of resistance R_1 is switched on, the kettle begins to boil tea in 6 minutes. When second coil of resistance R_2 is switched on, the boiling begins in 8 minutes. The value of $\mathrm{R}_1/\mathrm{R}_2$ is

(A) $\frac{7}{3}$

(B) $\frac{3}{7}$

(C) $\frac{3}{4}$

(D) $\frac{4}{3}$

Q.7

An electric fan and a heater are marked as $100\ \mathrm{W}, 220\ \mathrm{V}$ and $1000\ \mathrm{W}, 220\ \mathrm{V}$ respectively. The resistance of heater is

(A) equal to that of fan

(B) lesser than that of fan

(C) greater than that of fan

(D) zero

Q.8

Water boils in the electric kettle in 15 minutes after switching on. If the length of heating wire is decreased to $2/3$ of its initial value, then the same amount of water will boil with the same supply voltage in

(A) 8 minutes

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(B) 10 minutes

(C) 12 minutes

(D) 15 minutes

Q.9

It takes 12 minutes to boil 1 litre of water in an electric kettle. Due to some defect it becomes necessary to remove 20% turns of heating coil of the kettle. After repair, how much time will it take to boil 1 litre of water ?

(A) 9.6 minute

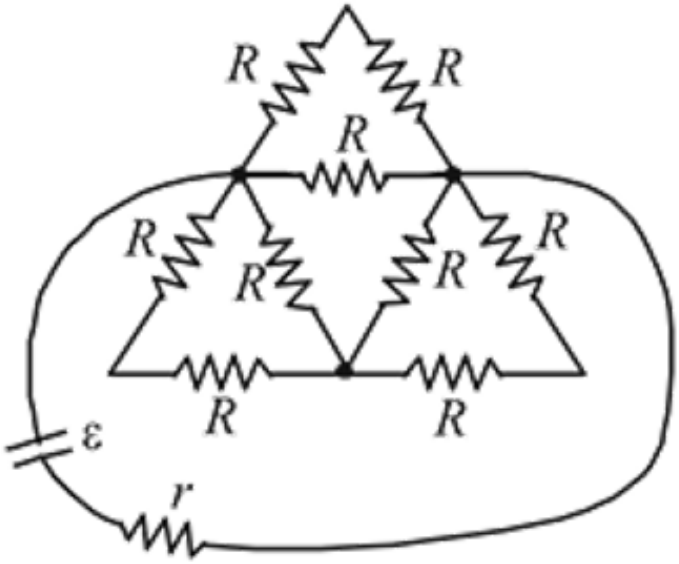
(B) 14.4 minute

(C) 16.8 minute

(D) 18.2 minute

Q.10

For maximum power from battery the internal resistance of battery r is



(A) $10R$

(B) $\frac{4R}{9}$

(C) $\frac{R}{8}$

(D) $\frac{10R}{9}$

Q.11

Two cities are $150\ \text{km}$ apart. Electric power is sent from one city to another city through copper wires. The fall of potential per km is 8 volt and the average resistance per km is 0.5Ω. The power loss in the wires is:

(A) $19.2\ \text{W}$

(B) $19.2\ \text{kW}$

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(C) $19.2\ \text{J}$

(D) $12.2\ \text{kW}$

Q.12

The charge flowing through a resistance R varies with time t as $Q = at - bt^2$. The total heat produced in R by the time current ceases is

(A) $a^3R/6b$

(B) $a^3R/3b$

(C) $\frac{a^3R}{2b}$

(D) $\frac{a^3R}{b}$

Q.13

Two cells with the same emf E and different internal resistances r_1 and r_2 are connected in series to an external resistance R. The value of R so that the potential difference across the first cell be zero is

(A) $\sqrt{r_1 r_2}$

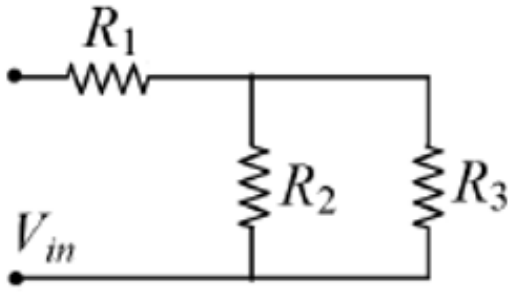
(B) $r_1 + r_2$

(C) $r_1 - r_2$

(D) $\frac{r_1 + r_2}{2}$

Q.14

For ensuring dissipation of same energy in all three resistors (R_1, R_2, R_3) connected as shown in figure, their values must be related as



(A) $R_1 = R_2 = R_3$

(B) $R_2 = R_3$ and $R_1 = 4R_2$

(C) $R_2 = R_3$ and $R_1 = \frac{1}{4}R_2$

(D) $R_1 = R_2 + R_3$

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Q.15

A coil of wire of resistance 50Ω is embedded in a block of ice. If a potential difference of $210\ \mathrm{V}$ is applied across the coil, the amount of ice melted per second will be (Latent heat of fusion of ice $= 80\mathrm{cal/gm}$)

(A) $4.12\mathrm{gm}$

(B) $4.12\ \mathrm{kg}$

(C) $3.68\ \mathrm{kg}$

(D) $2.625\mathrm{gm}$

Answers & Solutions

Q.1 Answer:

$R = r$

Solution:

The current in the circuit is

$I = \frac{E}{(R+r)}$

Therefore the power output of the device is given by

$P = I^2R = \frac{E^2R}{(R+r)^2}$ **(1)**

For given values of E and r, power output P will be maximum if $dP/dR = 0$ and $d^2P/dR^2 < 0$. Differentiating (i) with respect to R we get (with E and r fixed)

$\frac{dP}{dR} = \frac{E^2}{(R+r)^2}\left\{1 - \frac{2R}{(R+r)}\right\}$

Now $dP/dR = 0$ **if**

$1 - \frac{2R}{(R+r)} = 0$

which gives $R = r$. Thus, P will be either maximum or minimum when $R = r$. To decide whether P is maximum at $R = r$, we find d^2P/dR^2 at $R = r$. If its value is negative, P will be maximum. Differentiating (ii) we have

$$\frac{d^2P}{dR^2} = \frac{2E^2}{(R+r)^3}\left\{\frac{3R}{(R+r)} - 2\right\}$$

$$\therefore \left(\frac{d^2P}{dR^2}\right)_{\text{at } R=r} = -\frac{E^2}{8r^3}$$

which is negative. Hence P is maximum when $R = r$.

Q.2 Answer:

$6\ \mathrm{min}$

Solution:

Let H be the amount of heat energy needed to boil the given quantity of water. If R_1 and R_2 are the resistances of coils and V is the applied voltage, then

$H = \frac{V^2t_1}{R_1} = \frac{V^2t_2}{R_2}$

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or $\frac{R_2}{R_1}=\frac{t_2}{t_1}=\frac{15}{10}=\frac{3}{2}$

When the coils are connected in parallel, the combined resistance is given by

$\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}=\frac{R_1+R_2}{R_1R_2}$

Or $R=\frac{R_1R_2}{R_1+R_2}$

If the water takes t **minutes to boil, then**

$\frac{V^2t}{R}=\frac{V^2t_1}{R_1}$

$$t=t_1\frac{R}{R_1}=t_1\times\frac{R_1R_2}{(R_1+R_2)}\cdot\frac{1}{R_1}$$

$$\text{or }\ =t_1\times\frac{1}{\left(\frac{R_1}{R_2}+1\right)}=\frac{10\times1}{\left(\frac{2}{3}+1\right)}=6\text{ min}$$

Hence the correct choice is (a).

Q.3 Answer:

When the coil is divided into four equal parts and all the four parts are connected to the battery in parallel

Solution:

(c) $\mathrm{R}=\frac{\rho\ell}{\mathrm{A}}$

When wire is cut into 4 pieces and connected in parallel. $R_{\text{eff.}}=\frac{R}{16}\Rightarrow P_C=16P$

$P_A:P_B:P_C:P_D::\frac{V^2}{R}:\frac{V^2}{R/4}:\frac{V^2}{R/16}:\frac{V^2}{R/2}$

Q.4 Answer:

5%

Solution:

Resistance of bulb is constant

$P=\frac{V^2}{R}\Rightarrow\frac{\mathrm{D}p}{p}=\frac{2\mathrm{D}V}{V}+\frac{\mathrm{D}R}{R}$

$\frac{\mathrm{D}p}{p}=2\times2.5+0=5\%$

Q.5 Answer:

87.5%

Solution:

Efficiency is given by $\eta=\frac{\text{output}}{\text{input}}=\frac{5\times15\times14}{10\times8\times15}=0.875$ **or** 87.5%

Q.6 Answer:

$\frac{3}{4}$

Solution:

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$$\text{Heat supplied } = \frac{V^2}{R} \times t$$

$$\Rightarrow \frac{t_1}{R_1} = \frac{t_2}{R_2} \Rightarrow \frac{6}{R_1} = \frac{8}{R_2} \Rightarrow \frac{R_1}{R_2} = \frac{3}{4}$$

Q.7 Answer:

lesser than that of fan

Solution:

As $R \propto V^2/P$ or $R \propto 1/P$ so resistance of heater is less than that of fan.

Q.8 Answer:

10 minutes

Solution:

$H = \frac{V^2}{R_2} \times 15 \times 60 = \frac{V^2}{(2/3)R} \times t$
or $t = \frac{2}{3} \times 15 \times 60 = 600\ s = 10$ **minutes.**

Q.9 Answer:

9.6 minute

Solution:

As voltage of appliance remains constant, the amount of heat produced is given by,

$$H = \frac{V^2}{R}t$$

when resistance is reduced by 20%, new resistance is $R' = R - 0.2R = 0.8R$

$$H' = \frac{V^2}{0.8R}t'$$

Equating Eqs. (i) and (ii), we get

$$\frac{V^2}{R}t = \frac{V^2}{0.8R}t'$$

$$\Rightarrow t' = 0.8t = 0.8 \times 12 = 9.6 \text{ min}$$

Q.10 Answer:

$\frac{4R}{9}$

Solution:

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No solution available

Q.11 Answer:

$19.2\ \mathrm{kW}$

Solution:

Total resistance $\mathrm{R} = (0.5\Omega/\mathrm{km}) \times (150\ \mathrm{km})$
$= 75\Omega$
Total voltage drop $= (8\ \mathrm{V/km}) \times (150\ \mathrm{km})$
$= 1200\ \mathrm{V}$
Power loss $= \frac{(\Delta \mathrm{V})^2}{\mathrm{R}} = \frac{(1200)^2}{75}\ \mathrm{W}$
$= 19200\ \mathrm{W} = 19.2\ \mathrm{kW}$

Q.12 Answer:

$a^3R/6b$

Solution:

$Q = at - bt^2 \quad i = \frac{dQ}{dt} = a - 2bt$
$i = 0$ **for** $t = t_0 = a/2b$**, i.e., current flow from** $t = 0$ **to** $t = t_0$
The heat produced $= \int_0^{l_0} i^2 R dt$**. Putting the value of** i **we get heat produced** $= \frac{a^3R}{6b}$**.**

Q.13 Answer:

$r_1 - r_2$

Solution:

Current in the circuit is $I = \frac{2E}{R+r_1+r_2}, \quad V_1 = E - I_1 = 0$
$\Rightarrow \quad E - \frac{2Er_1}{R+r_1+r_2} = 0 \Rightarrow R = r_1 - r_2$
$\therefore$
(c)

Q.14 Answer:

$R_2 = R_3$ and $R_1 = \frac{1}{4}R_2$

Solution:

As the voltage in R_2 **and** R_3 **is same**
therefore, according to,

$$H = \frac{V^2}{R} \cdot t, \quad R_2 = R_3$$

Also the energy in all resistance is
same.

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$\therefore$

$i^2 R_1 t = i_1^2 R_2 t$

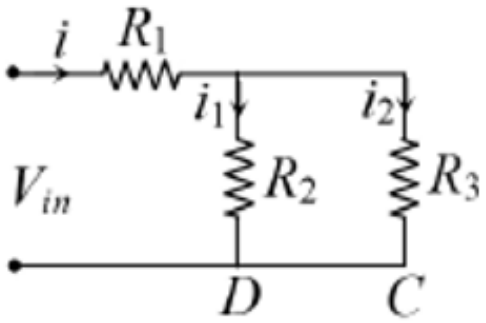
Using $i_1 = \frac{R_3}{R_2+R_3} i = \frac{R_3}{R_3+R_3} i = \frac{1}{2} i$

Thus, $i^2 R_1 t = \frac{i^2}{4} R_2 t$ **or,**

$R_1 = \frac{R_2}{4}$

$\therefore$

(c)



Q.15 Answer:

2.625gm

Solution:

$\frac{Q}{t} = \frac{V^2}{4.2R} = \frac{m}{t} L; \quad \frac{m}{t} = \frac{V^2}{4.2RL} = \frac{(210)^2}{4.2 \times 50 \times 80} = 2.625\text{gm}$

(d)

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